

1. BINARY LATTICES

For positive integers m and n , an (m, n) *lattice* is a rectangular grid that has $m + 1$ rows and $n + 1$ columns of integers, which we call *vertices*. All our lattices will be *binary*, meaning that each vertex is 0 or 1. A *cell* is a $(1, 1)$ binary lattice, so that an (m, n) binary lattice is made up of mn cells. The sixteen possible cells are

$\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}, \begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}, \begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}, \begin{smallmatrix} 0 & 0 \\ 1 & 1 \end{smallmatrix}, \begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}, \begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}, \begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}, \begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}, \begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}, \begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}, \begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}, \begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}, \begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}, \begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}, \begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix},$ and $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$.

We further define that a lattice is *framed* if it is binary and every vertex on the boundary of the lattice is 0, and a lattice is *proper* if it is framed and does not contain any $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ or $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ cells. For example, the right side of Figure ?? is a $(5, 4)$ proper lattice.